

1 Some discrete probability distributions

Binomial distribution

$$f(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, \dots, n.$$

$$E(X) = np, \quad \text{Var}(X) = np(1-p).$$

Poisson distribution

$$f(x) = \frac{\mu^x}{x!} e^{-\mu}, \quad x = 0, 1, 2, \dots$$

$$E(X) = \mu, \quad \text{Var}(X) = \mu.$$

Hypergeometric distribution

$$f(x) = \frac{\binom{k}{x} \binom{N-k}{n-x}}{\binom{N}{n}}, \quad x = 0, 1, \dots, \min(k, n).$$

$$E(X) = \frac{nk}{N}, \quad \text{Var}(X) = \frac{N-n}{N-1} \cdot n \cdot \frac{k}{N} \cdot \left(1 - \frac{k}{N}\right).$$

Geometric distribution

$$f(x) = p(1-p)^{x-1}, \quad x = 1, 2, \dots$$

$$E(X) = \frac{1}{p}, \quad \text{Var}(X) = \frac{1-p}{p^2}.$$

Negative binomial distribution

$$f(x) = \binom{x-1}{k-1} p^k (1-p)^{x-k}, \quad x = k, k+1, \dots$$

$$E(X) = \frac{k}{p}, \quad \text{Var}(X) = k \frac{1-p}{p^2}.$$

Special case: $k = 1$ gives the geometric distribution.

Discrete uniform distribution

$$f(x) = \frac{1}{k}, \quad x = x_1, \dots, x_k.$$

$$E(X) = \frac{1}{k} \sum_{i=1}^k x_i, \quad \text{Var}(X) = \frac{1}{k} \sum_{i=1}^k (x_i - E(X))^2.$$

Multinomial distribution

$$f(x_1, \dots, x_k; p_1, \dots, p_k, n) = \frac{n!}{x_1! \dots x_k!} p_1^{x_1} \dots p_k^{x_k}, \text{ where } \sum_{i=1}^k x_i = n \text{ and } \sum_{i=1}^k p_i = 1.$$

$$E(X_i) = np_i, \quad \text{Var}(X_i) = np_i(1 - p_i), \quad \text{Cov}(X_i, X_j) = -np_i p_j \text{ for } i \neq j.$$

Special case: For $k = 2$ is X_1 binomially distributed.

2 Some continuous probability distributions

Normal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{1}{2} \frac{(x - \mu)^2}{\sigma^2}\right), \quad -\infty < x < \infty.$$

$$E(X) = \mu, \quad \text{Var}(X) = \sigma^2.$$

$$F(x) = P(X \leq x) = \Phi\left(\frac{x - \mu}{\sigma}\right).$$

Multivariate normal distribution

$$f(x_1, \dots, x_n) = \frac{1}{(2\pi)^{n/2} |\Sigma|^{1/2}} \exp\{-0.5(\mathbf{x} - \boldsymbol{\mu})^T \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu})\}$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}, \quad \boldsymbol{\mu} = \begin{bmatrix} \mu_1 \\ \vdots \\ \mu_n \end{bmatrix}, \quad \Sigma = \begin{bmatrix} \sigma_{11} & \dots & \sigma_{1n} \\ \vdots & \ddots & \vdots \\ \sigma_{n1} & \dots & \sigma_{nn} \end{bmatrix}$$

Exponential distribution

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x \geq 0.$$

$$E(X) = \beta, \quad \text{Var}(X) = \beta^2.$$

Comment: Often common to use the parameter $\lambda = 1/\beta$.

Lognormal distribution

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma x} \exp\left(-\frac{1}{2} \frac{(\ln(x) - \mu)^2}{\sigma^2}\right), \quad x > 0.$$

$$E(X) = e^{\mu + \sigma^2/2}, \quad \text{Var}(X) = e^{2(\mu + \sigma^2)} - e^{2\mu + \sigma^2}.$$

χ^2 -distribution (Chi-Squared distribution)

$$f(x) = \frac{1}{2^{\nu/2}\Gamma(\nu/2)} x^{\nu/2-1} e^{-x/2}, \quad x \geq 0, \quad \nu = 1, 2, \dots$$

$$E(X) = \nu, \quad \text{Var}(X) = 2\nu.$$

The parameter ν is often called degrees of freedom.

Comment: If $X_1, \dots, X_n$ are independent and have a normal distribution with expectation μ and variance σ^2 we have that

$$\begin{aligned} \sum_{i=1}^n \frac{(X_i - \mu)^2}{\sigma^2} & \text{ is } \chi^2\text{-distributed with } n \text{ degrees of freedom,} \\ \sum_{i=1}^n \frac{(X_i - \bar{X})^2}{\sigma^2} & \text{ is } \chi^2\text{-distributed with } n - 1 \text{ degrees of freedom,} \\ \sum_{i=1}^n (X_i - \bar{X})^2 & \text{ and } \bar{X} \text{ are independent.} \end{aligned}$$

t -distribution (Student t -distribution)

$$f(x) = \frac{\Gamma[(\nu+1)/2]}{\Gamma(\nu/2)\sqrt{\pi\nu}} \left(1 + \frac{x^2}{\nu}\right)^{-(\nu+1)/2}, \quad -\infty < x < \infty.$$

$$E(X) = 0 \text{ if } \nu \geq 2, \quad \text{Var}(X) = \frac{\nu}{\nu-2} \text{ if } \nu \geq 3.$$

Special case: $\nu = \infty$ gives the Normal distribution.

Comment: If Z has a standard normal distribution and V has a χ^2 -distribution with ν degrees of freedom and Z and V are independent, we have that

$$\frac{Z}{\sqrt{V/\nu}} \text{ is } t\text{-distributed with } \nu \text{ degrees of freedom.}$$

In particular this gives that if $X_1, \dots, X_n$ are independent and have a normal distribution with expectation μ and variance σ^2 we have that

$$\frac{\bar{X} - \mu}{S/\sqrt{n}} \text{ is } t\text{-distributed with } n - 1 \text{ degrees of freedom,}$$

$$\text{where } S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Gamma distribution

$$f(x) = \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, \quad x \geq 0.$$

$$E(X) = \alpha\beta, \quad \text{Var}(X) = \alpha\beta^2.$$

Special cases: $\alpha = 1$ gives the exponential distribution.

$\alpha = \nu/2$ and $\beta = 2$ give the χ^2 -distribution.

Weibull distribution

$$f(x) = \alpha\beta x^{\beta-1} e^{-\alpha x^\beta}, \quad x \geq 0.$$

$$E(X) = \alpha^{-1/\beta} \Gamma(1 + \frac{1}{\beta}), \quad \text{Var}(X) = \alpha^{-2/\beta} \left[\Gamma(1 + \frac{2}{\beta}) - \Gamma(1 + \frac{1}{\beta})^2 \right].$$

Special case: $\beta = 1$ gives the exponential distribution.

Continuous uniform distribution

$$f(x) = \frac{1}{b-a}, \quad a \leq x \leq b.$$

$$E(X) = \frac{(a+b)}{2}, \quad \text{Var}(X) = \frac{(b-a)^2}{12}.$$

Beta distribution

$$f(x) = \frac{1}{B(\alpha, \beta)} x^{\alpha-1} (1-x)^{\beta-1}, \quad 0 \leq x \leq 1.$$

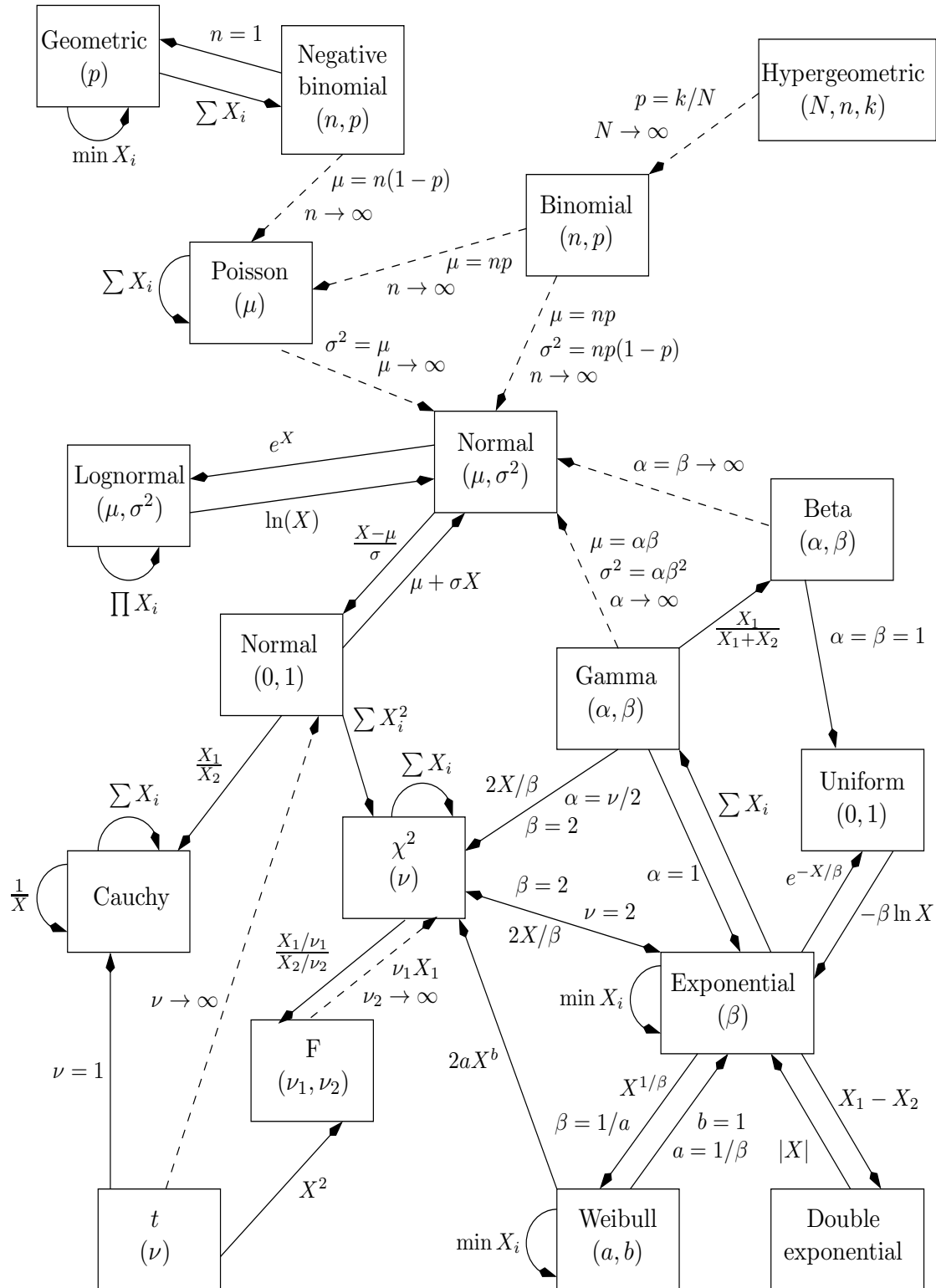
$$E(X) = \frac{\alpha}{\alpha + \beta}, \quad \text{Var}(X) = \frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}.$$

Comment: $B(\alpha, \beta) = \frac{\Gamma(\alpha)\Gamma(\beta)}{\Gamma(\alpha+\beta)}.$

Pareto distribution

$$f(x) = \frac{\beta\alpha^\beta}{x^{\beta+1}}, \quad \alpha < x < \infty.$$

$$E(X) = \frac{\beta\alpha}{\beta-1} \text{ if } \beta > 1, \quad \text{Var}(X) = \frac{\beta\alpha^2}{(\beta-1)^2(\beta-2)} \text{ if } \beta > 2.$$



Relationship among distributions. Solid lines represents transformations and special cases, while dashed lines represents limits. Adapted from Leemis, L. M. (1986). Relationships Among Common Univariate Distributions, *The American Statistician* **40**: 143-146.

3 Some results and definitions

The addition rule

$$\begin{aligned} P(A_1 \cup \dots \cup A_n) &= \sum_{i=1}^n P(A_i) - \sum_{i < j} P(A_i \cap A_j) + \sum_{i < j < k} P(A_i \cap A_j \cap A_k) - \dots \\ &\quad + (-1)^{n+1} P(A_1 \cap \dots \cap A_n). \end{aligned}$$

The multiplication rule

$$P(A_1 \cap \dots \cap A_n) = P(A_1)P(A_2|A_1) \cdots P(A_n|A_1 \cap \dots \cap A_{n-1}).$$

The rule of total probability

Let $B_1, \dots, B_k$ be a partition of the sample space. Then

$$P(A) = \sum_{i=1}^k P(A|B_i)P(B_i).$$

Bayes' rule

$$P(B_r|A) = \frac{P(B_r \cap A)}{P(A)} = \frac{P(A|B_r)P(B_r)}{\sum_{i=1}^k P(A|B_i)P(B_i)}$$

where $B_1, \dots, B_k$ is a partition of the sample space.

The correlation coefficient between X and Y

$$\rho(X, Y) = \frac{\text{Cov}(X, Y)}{\sqrt{\text{Var}(X)\text{Var}(Y)}} = \frac{\text{E}(XY) - \text{E}(X)\text{E}(Y)}{\sqrt{\text{Var}(X)}\sqrt{\text{Var}(Y)}}.$$

The central limit theorem

Let $X_1, X_2, \dots, X_n$ be a sequence of independent identically distributed random variables with $\text{E}(X_i) = \mu$ and $0 < \text{Var}(X_i) = \sigma^2 < \infty$. Then

$$\frac{\sqrt{n}(\bar{X} - \mu)}{\sigma} \xrightarrow{d} Z$$

where Z has a standard normal distribution.

Consistent estimator

An estimator $\hat{\theta}$ for a parameter θ is said to be consistent if $\hat{\theta} \xrightarrow{p} \theta$.

An estimator will be consistent if $E(\hat{\theta}) \rightarrow \theta$ and $\text{Var}(\hat{\theta}) \rightarrow 0$ as the number of observations goes to infinity.

The gamma function

Definition: $\Gamma(\alpha) = \int_0^{\infty} t^{\alpha-1} e^{-t} dt$, $\alpha > 0$.

$$\Gamma(\alpha) = (\alpha - 1)\Gamma(\alpha - 1), \quad \Gamma(n + 1) = n!, \quad n = 0, 1, 2, \dots$$

$$\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}, \quad \Gamma\left(\frac{2k+1}{2}\right) = \frac{2k-1}{2} \cdot \frac{2k-3}{2} \cdots \frac{1}{2} \sqrt{\pi}, \quad k = 1, 2, \dots$$

Some series

$$\sum_{k=0}^n \binom{n}{k} a^k b^{n-k} = (a + b)^n, \quad \sum_{k=0}^{\infty} \frac{a^k}{k!} = e^a, \quad \sum_{k=0}^n a^k = \frac{1 - a^{n+1}}{1 - a}.$$

$$\sum_{k=0}^{\infty} k a^k = \frac{a}{(1 - a)^2}, \quad \sum_{k=1}^n k = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}.$$

$$\text{If } 0 < a < 1, \text{ then: } \sum_{k=0}^{\infty} a^k = \frac{1}{1 - a}$$

4 Some results for functions of random variables.

The transformation formula

Let $Y = u(X)$, where the function u is strictly monotone and differentiable for all possible X . Let $X = u^{-1}(Y) = w(Y)$. If X has a continuous distribution, Y will also have a continuous distribution. Let $f(x)$ be the probability density function of X and let $g(y)$ be the probability density function of Y . We then have the relationship

$$g(y) = f(w(y)) \cdot |w'(y)|.$$

Some transformations

If X has a Weibull distribution with parameters α and β , then $Y = \alpha X^\beta$ has an exponential distribution with parameter 1.

If X has a normal distribution with expectation μ and variance σ^2 , then $Y = (X - \mu)/\sigma$ has normal distribution with expectation 0 and variance 1.

If X has a normal distribution with expectation μ and variance σ^2 , then $Y = (X - \mu)^2/\sigma^2$ has a χ^2 -distribution with 1 degree of freedom.

If X has a gamma distribution with parameters α and β , then $Y = 2X/\beta$ has a χ^2 -distribution with 2α degrees of freedom.

The distribution of some linear combinations

Let $X_1, \dots, X_n$ be independent variables.

If X_i has a normal distribution with expectation μ_i and variance σ_i^2 , then $Y = \sum_{i=1}^n a_i X_i$ has a normal distribution with expectation $\sum_{i=1}^n a_i \mu_i$ and variance $\sum_{i=1}^n a_i^2 \sigma_i^2$.

If X_i has a binomial distribution with parameters m_i and p , then $Y = \sum_{i=1}^n X_i$ has a binomial distribution with parameters $\sum_{i=1}^n m_i$ and p .

If X_i has a Poisson distribution with parameter μ_i then $Y = \sum_{i=1}^n X_i$ has a Poisson distribution with parameter $\sum_{i=1}^n \mu_i$.

If X_i has a gamma distribution with parameters α_i and β , then $Y = \sum_{i=1}^n X_i$ has a gamma distribution parameters $\sum_{i=1}^n \alpha_i$ and β .

If X_i has an exponential distribution with expectation β , then $Y = \sum_{i=1}^n X_i$ has a gamma distribution parameters n and β .

If X_i has a χ^2 -distribution with ν_i degrees of freedom, then $Y = \sum_{i=1}^n X_i$ has a χ^2 -distribution with $\sum_{i=1}^n \nu_i$ degrees of freedom.

Expectation and variance of linear combinations

Let $Y = \sum_{i=1}^n a_i X_i + b$. Then

$$E(Y) = \sum_{i=1}^n a_i E(X_i) + b,$$

$$\text{Var}(Y) = \sum_{i=1}^n a_i^2 \text{Var}(X_i) + 2 \sum_{i=1}^n \sum_{j=1}^{i-1} a_i a_j \text{Cov}(X_i, X_j).$$

Expectation of functions

Discrete variables:

$$E[g(X_1, \dots, X_n)] = \sum_{x_1} \cdots \sum_{x_n} g(x_1, \dots, x_n) f(x_1, \dots, x_n).$$

Continuous variables:

$$E[g(X_1, \dots, X_n)] = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} g(x_1, \dots, x_n) f(x_1, \dots, x_n) dx_1 \cdots dx_n.$$

5 Maximum likelihood estimators

Maximum likelihood estimators $\hat{\theta}$ are under certain regularity conditions (for instance that the range of possible values of X should not depend on θ) invariant, consistent, asymptotically unbiased and asymptotically normally distributed. The variance of a maximum likelihood estimator can be approximated from the second derivative of the log likelihood function:

$$\text{Var}(\hat{\theta}) \approx \frac{1}{J(\theta)} \quad \text{or} \quad \text{Var}(\hat{\theta}) \approx \frac{1}{E(J(\theta))} \quad \text{where} \quad J(\theta) = -\frac{\partial^2}{\partial \theta^2} l(\theta).$$

In particular it holds that:

$$\frac{\hat{\theta} - \theta}{\sqrt{1/J(\hat{\theta})}} \approx N(0, 1).$$

6 Hazard rate

The hazard rate for a continuous random variable is found by:

$$r(t) = \frac{f(t)}{1 - F(t)}$$

7 Derivatives and integrals

Derivatives

function	derivative
$y = x^n$	$y' = nx^{n-1}$
$y = e^x$	$y' = e^x$
$y = e^{u(x)}$	$y' = u'(x)e^{u(x)}$
$y = \ln(x)$	$y' = \frac{1}{x}$
$y = a^x$	$y' = a^x \ln(a)$
$y = u(x) \pm v(x)$	$y' = u'(x) \pm v'(x)$
$y = u(x) \cdot v(x)$	$y' = u'(x) \cdot v(x) + u(x) \cdot v'(x)$
$y = f[u(x)]$	$y' = f'[u(x)]u'(x)$

Integrals

$$\begin{aligned}\int x^n dx &= \frac{x^{n+1}}{n+1} + C, & n \neq -1 \\ \int \frac{1}{x} dx &= \ln(|x|) + C \\ \int [u(x) \pm v(x)] dx &= \int u(x) dx \pm \int v(x) dx \\ \int e^{ax} dx &= \frac{1}{a} e^{ax} + C \\ \int x e^{ax} dx &= \frac{e^{ax}}{a^2} (ax - 1) + C \\ \int x^2 e^{ax} dx &= e^{ax} \left(\frac{x^2}{a} - \frac{2x}{a^2} + \frac{2}{a^3} \right) + C \\ \int u'(x) e^{u(x)} dx &= e^{u(x)} + C\end{aligned}$$

Integration by parts:

$$\int_a^b u(x) \cdot v'(x) dx = [u(x) \cdot v(x)]_a^b - \int_a^b u'(x) \cdot v(x) dx$$

8 The natural logarithm

$$\begin{aligned}\ln(a \cdot b) &= \ln(a) + \ln(b), & \ln\left(\frac{a}{b}\right) &= \ln(a) - \ln(b) \\ \ln(a^b) &= b \ln(a), & \ln(e^a) &= a\end{aligned}$$

9 Markov chain Monte Carlo

Metropolis Hastings algorithm (MH)

Suppose the proposal mechanism has density $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$, and an initial state $\boldsymbol{\theta}_0$ is given. Then the MH algorithm is given by

1. set $t \leftarrow 1$
2. simulate $\boldsymbol{\theta}^* \sim q(\cdot|\boldsymbol{\theta}_{t-1})$.
3. calculate accept probability

$$\alpha = \min \left(1, \frac{q(\boldsymbol{\theta}_{t-1}|\boldsymbol{\theta}^*)g(\boldsymbol{\theta}^*)}{q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})g(\boldsymbol{\theta}_{t-1})} \right)$$

4. simulate $u \sim \text{uniform}(0,1)$.
5. if $u < \alpha$ set $\boldsymbol{\theta}_t = \boldsymbol{\theta}^*$. Otherwise set $\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1}$.
6. set $t \leftarrow t + 1$ and go back to step 2.

Random walk Metropolis Hastings (RWMH): If $\boldsymbol{\theta}^* \sim N(\boldsymbol{\theta}_{t-1}, \Sigma)$ and $q(\boldsymbol{\theta}_{t-1}|\boldsymbol{\theta}^*) = q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$ then we have a RWMH algorithm.

Independent MH sampler: If we let $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1})$ be independent of $\boldsymbol{\theta}_{t-1}$, i.e. $q(\boldsymbol{\theta}^*|\boldsymbol{\theta}_{t-1}) = r(\boldsymbol{\theta}^*)$ for some pdf $r(\boldsymbol{\theta}^*)$.

Gibbs Sampler

Let the state $\boldsymbol{\theta}$ be partitioned into two (or more) blocks $\boldsymbol{\theta} = (\boldsymbol{\theta}^{(1)}, \boldsymbol{\theta}^{(2)})$. Then alternate between updating the blocks according to the **conditional** distributions:

- Input: old state $\boldsymbol{\theta}_{t-1} = (\boldsymbol{\theta}_{t-1}^{(1)}, \boldsymbol{\theta}_{t-1}^{(2)})$
- Update $\boldsymbol{\theta}_t^{(1)} \sim \pi(\boldsymbol{\theta}^{(1)}|\boldsymbol{\theta}_{t-1}^{(2)})$.
- Update $\boldsymbol{\theta}_t^{(2)} \sim \pi(\boldsymbol{\theta}^{(2)}|\boldsymbol{\theta}_t^{(1)})$
- Return new state $\boldsymbol{\theta}_t = (\boldsymbol{\theta}_t^{(1)}, \boldsymbol{\theta}_t^{(2)})$

10 Rules for recognizing distributions

- **Normal distribution**

$$\log p(x) = a + bx + cx^2 \Rightarrow x \sim N\left(-\frac{b}{2c}, -\frac{1}{2c}\right)$$

- **Exponential distribution**

$$\log p(x) = a + bx, \ x > 0 \Rightarrow x \sim \text{Exp}(-1/b)$$

- **Gamma distribution**

$$\log p(x) = a + b \log(x) + cx, \ x > 0 \Rightarrow x \sim \text{Gamma}(b+1, -1/c)$$

- **Beta distribution**

$$\log p(x) = a + b \log(x) + c \log(1-x), \ 0 < x < 1 \Rightarrow x \sim \text{Beta}(b+1, c+1)$$

- **Inverse Gamma distribution**

$$\log p(x) = a + b \log(x) - c/x, \ x > 0 \Rightarrow x \sim \text{InverseGamma}(-(b+1), c)$$

- **Multivariate Normal distribution** (**b** vector; **c** symmetric positive definite matrix)

$$\log p(\mathbf{x}) = a + \mathbf{b}^T \mathbf{x} + \frac{1}{2} \mathbf{x}^T \mathbf{c} \mathbf{x} \Rightarrow \mathbf{x} \sim N(-\mathbf{c}^{-1} \mathbf{b}, -\mathbf{c}^{-1})$$